

The Atmosphere

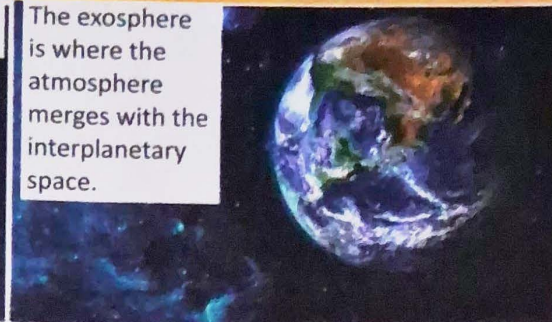


Before Reading

Auroras, the moving coloured lights, occur in the thermosphere.



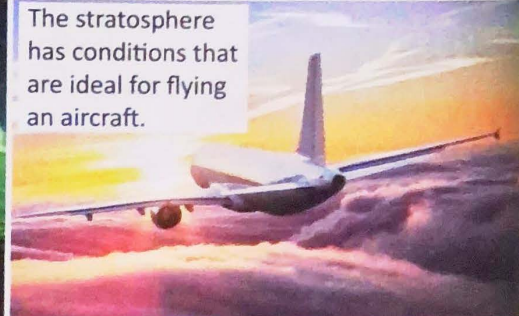
The exosphere is where the atmosphere merges with the interplanetary space.



The **meteors** burn up in the mesosphere.



The stratosphere has conditions that are ideal for flying an aircraft.



The troposphere is where changes in weather conditions occur.



Making Connection



Learning Outcomes

At the end of this lesson, learners will be able to:

- **understand** the importance and composition of the atmosphere.
- **draw and label** the structure of the atmosphere.
- **comprehend** how layers of the atmosphere sustain life.
- **understand** the meaning of and causes behind the greenhouse effect.
- **discuss** the causes of and solutions for global warming.
- **analyse** the challenges faced because of the depletion of the ozone layer.

The atmosphere with its layer of gases make the Earth unique among all the other planets of the solar system.

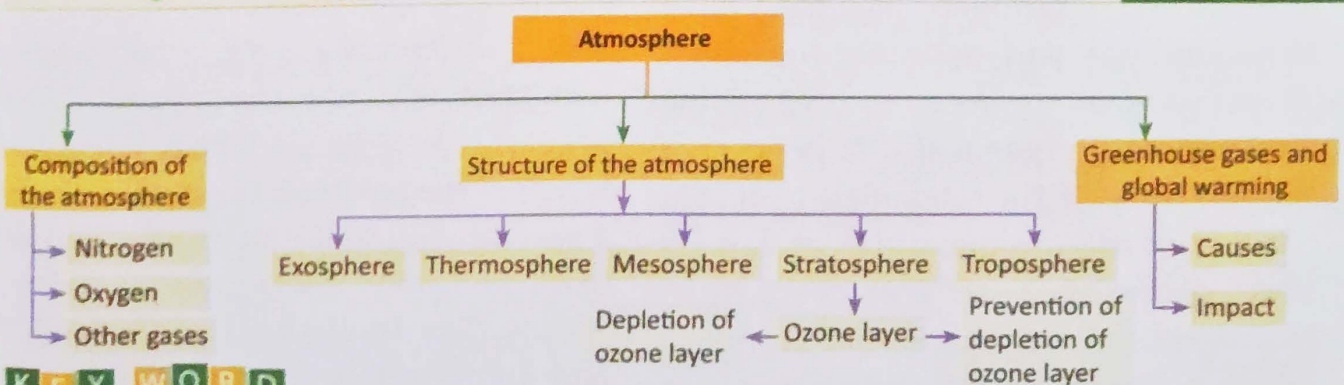
- Where is the ozone layer located? How does the ozone layer protect the Earth?

21st
CENTURY
SKILLS

- information literacy
- critical thinking

During Reading

Visual Mapping



KEY WORD

meteor: a small rocky celestial body that enters the Earth's atmosphere, starts glowing as a result of friction with the atmosphere and appears as a streak of light. It is also known as a shooting star.



Warm-Up

Some environmental problems and ways to save the world are given below. Colour code the environmental problems as ■ and ways to save the world as ■.



factory ☐



water pollution ☐



solar energy ☐



recycle ☐



deforestation ☐



waste segregation ☐



car smoke ☐



do not burn ☐



plant a tree ☐



use a bike ☐

WHAT IS ATMOSPHERE?

The atmosphere is made up of a mixture of gases that form a layer surrounding the Earth. The atmosphere is held in place by the gravitational force of our planet.

It forms the vital gaseous realm of the Earth. Although, the atmosphere extends beyond 1500 km above the Earth's surface, 99 per cent of its mass lies within a height of 32 km from the ground. This is because the density of the atmosphere is greater in the lower layers, as the upper layers compress them into a dense mass.

varying amounts of water vapour, and suspended solid particles such as dust, salt and pollen.

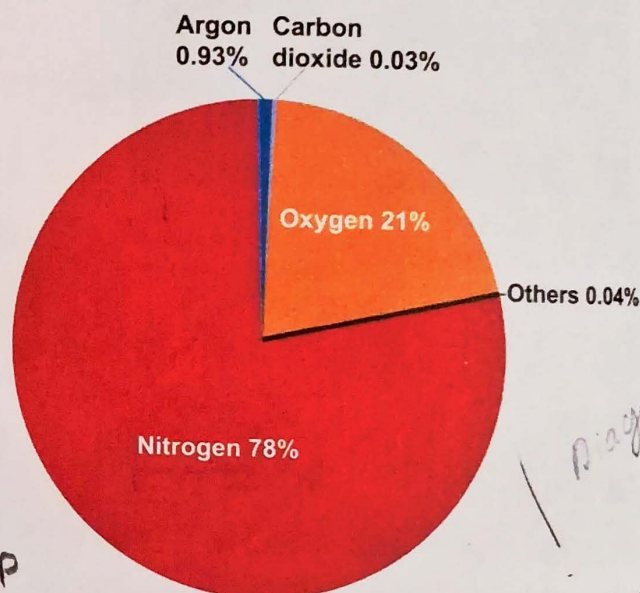


Fig. 2.1: composition of the atmosphere

COMPOSITION OF THE ATMOSPHERE

Fig. 2.1 shows the composition of the atmosphere. Nitrogen is the most abundant gas and accounts for about 78 per cent of the total volume of the atmosphere. Oxygen forms 21 per cent of the atmosphere. The remaining 1 per cent consists of minute quantities of various trace gases such as carbon dioxide, hydrogen, helium, ozone and argon. In its lower layers the atmosphere contains

STRUCTURE OF THE ATMOSPHERE

The differences in composition, temperature, density and other properties help us to identify five distinct layers of the atmosphere. Although there is no clear boundary separating one layer from the other, each layer is uniquely identifiable. The layers vary in thickness and are found at different heights from

the surface of the Earth. Refer to Fig. 2.2 as we read about the layers of the atmosphere.

Troposphere

Tropo = change

- The lowest layer of the atmosphere is known as the **troposphere**.
- The thickness of this layer is not uniform as its upper limit is higher (16 km) above the Equator than above the two poles (8 km).
- It is the **densest layer**, which contains 75 per cent of the total mass of the atmosphere.
- This layer contains the air we breathe. It also has almost all the dust particles and water vapour present in the atmosphere. All changes in the weather conditions involving clouds, rain, snow, fog and storm, occur in the troposphere due to the presence of water vapour.
- Both temperature and pressure of air decrease as its density gradually decreases, towards higher levels of the troposphere.
- This layer of the atmosphere acts as a blanket protecting the Earth from extreme heat during the day. It also traps **terrestrial heat**, thereby keeping the Earth warm during the night.

- The upper limit of the troposphere is marked by the **tropopause**, where temperature stops decreasing with height.

- Very-high-speed winds called **jet streams** blow in the tropopause. These winds are capable of influencing the surface winds on the Earth, thereby bringing a change in the weather. For example, the subtropical **westerly jet streams** influence the **western disturbances**, which cause rainfall in north-west India during the winter months.

→ no clouds in this area

Stratosphere

*lower layer has - - -
stratopause has - - -*

- Above the troposphere lies the **stratosphere**, extending from the tropopause till about 50 km from the Earth's surface.
- This layer of the atmosphere **provides ideal flying conditions** for aircraft. This is because the absence of water vapour makes this layer free from weather disturbances. Hence, **no turbulence** can be felt.
- Within the stratosphere lies a layer that has **rich concentration of ozone** and is known as **ozonosphere**. Ozone is vital for our survival as it absorbs the **harmful ultraviolet radiations** (UV radiations) emitted by the Sun. This

← G-R

*1) provides ideal conditions for a flight
2) crucial for survival of life on earth*

KEY WORDS

terrestrial heat: heat emitted by the Earth into the atmosphere

jet streams: high-speed winds of the upper atmosphere that influence and force the surface winds to blow in the direction they move.

ultraviolet: ultraviolet light is beyond the normal range of colours that human eyes can see. Too much ultraviolet light can be bad for your skin. It is often simply called UV rays.



Knowledge Box

In the troposphere, there is 1° decrease in temperature with every 165 metres of ascent. This is known as the **Normal Lapse Rate**. Along with temperature, the **density of air declines** and air becomes thinner away from the Earth's surface. This leads to a decline in the amount of oxygen and other vital gases. Hence, mountaineers carry oxygen cylinders when they climb great heights.

increases the temperature of this layer. However, excessive use of harmful **chlorofluorocarbon (CFC) gases** is gradually depleting this ozone layer, leading to health hazards.

- The upper limit of this layer is marked by the **stratopause**, where the temperature begins to fall with altitude again.

Mesosphere

- Resting above the stratosphere and lying at a height of around 50 km to 85 km from the Earth's surface is the layer of the middle atmosphere known as the **mesosphere**.
- This is the **coldest atmospheric layer** with the temperature dipping to -90° Celsius ($^\circ\text{C}$) in some parts. The temperature decreases with height throughout this layer.
- This layer is of **vital importance** because **meteors slow down and burn up** due to friction and frictional heat when they enter the mesosphere.

KEY WORD

chlorofluorocarbon gases: a gas used in refrigerators and in some aerosol containers. CFCs are not often used anymore because they damage the ozone layer of the Earth's atmosphere.

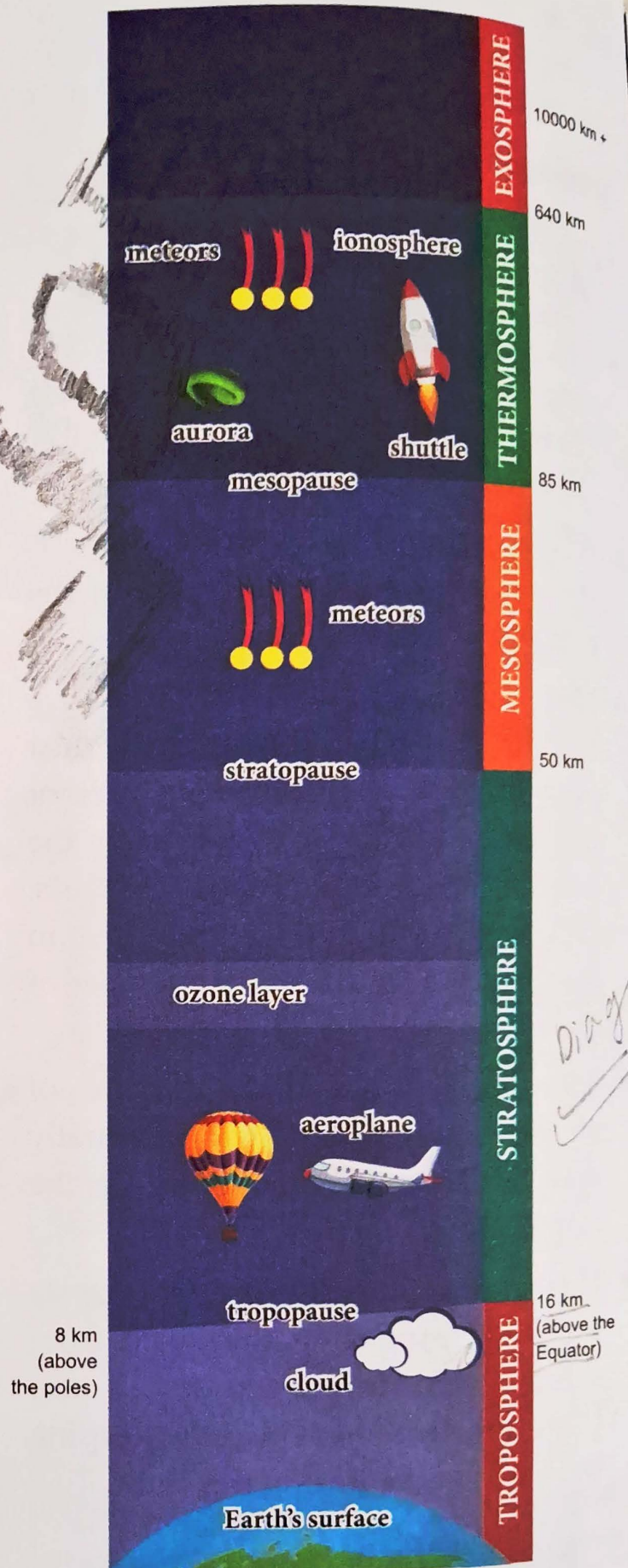


Fig. 2.2: structure of the atmosphere

- The mesopause marks the boundary between the mesosphere and the thermosphere, where the atmospheric temperature reaches its lowest point.

Thermosphere (Ionosphere)

- This layer extends to a height of around 640 km from the surface of the Earth.
- The temperature increases with height as this layer absorbs various types of solar radiations.



• creativity • collaboration
• communication • information literacy
• social and cross-cultural interaction

Know Your Geography

Thermosphere is where auroras occur. They are bright, beautiful bands of light that we sometimes see near the polar regions. They are caused by high-energy particles from the Sun interacting with the atoms in the ionosphere of the atmosphere.



Aurora borealis, also known as polar lights or northern lights

People around the world tell stories about the auroras. They are aurora borealis or northern lights in the northern hemisphere and aurora australis or southern lights in the southern hemisphere. Ask your family and friends to share stories about the aurora. You could even write or draw your own aurora story!

cross-curricular connect: Science and English
art-integrated learning

KEY WORD

rarefied: rarefied air does not contain much oxygen

- 4-R
- This results in the breaking up of the molecules of atmospheric gases into electrically charged particles called ions. These ions reflect radio waves back to the Earth and enable radio broadcasts and wireless communication. This layer is thus known as ionosphere.
 - As there are relatively few molecules and atoms in the thermosphere, even absorbing small amounts of solar energy can significantly increase the air temperature, making the thermosphere the hottest layer in the atmosphere.

Exosphere

- The exosphere is the uppermost layer of the atmosphere. It is thousands of kilometres thick and eventually merges with the interplanetary space.
- The air is extremely thin and rarefied, which makes the atmospheric pressure the lowest in this layer. It is composed of traces of very light gases, such as hydrogen, helium and oxygen.

IMPORTANCE OF THE ATMOSPHERE

(Due to lack of gravitational pull)

The presence of the atmosphere is one of the factors that makes life on the Earth possible.

1. The oxygen in the atmosphere is essential to humans for respiration, while carbon dioxide and nitrogen

JOE

are essential for plants. Water vapour in the atmosphere is responsible for various forms of precipitation, while ozone prevents the harmful ultraviolet rays from reaching the Earth's surface.

2. Incoming solar radiation heats up the Earth's surface during day time. At night, the Earth radiates back the heat, much of which is absorbed by the atmosphere. Greenhouse gases (GHGs) such as carbon dioxide and water vapour, trap the terrestrial heat and thus maintain an optimum temperature for all living beings.

3. Meteors burn up in the atmosphere before they can hit the Earth's surface.

GREENHOUSE EFFECT AND GLOBAL WARMING

As you already know, the rays of the Sun heat up the Earth's surface during the day. The Earth cools at night by releasing the heat in the form of terrestrial radiation. Some of this heat is absorbed by the GHGs of the atmosphere and does not escape into the outer space. This prevents the Earth from becoming excessively cold at night. This is known as **greenhouse effect**.

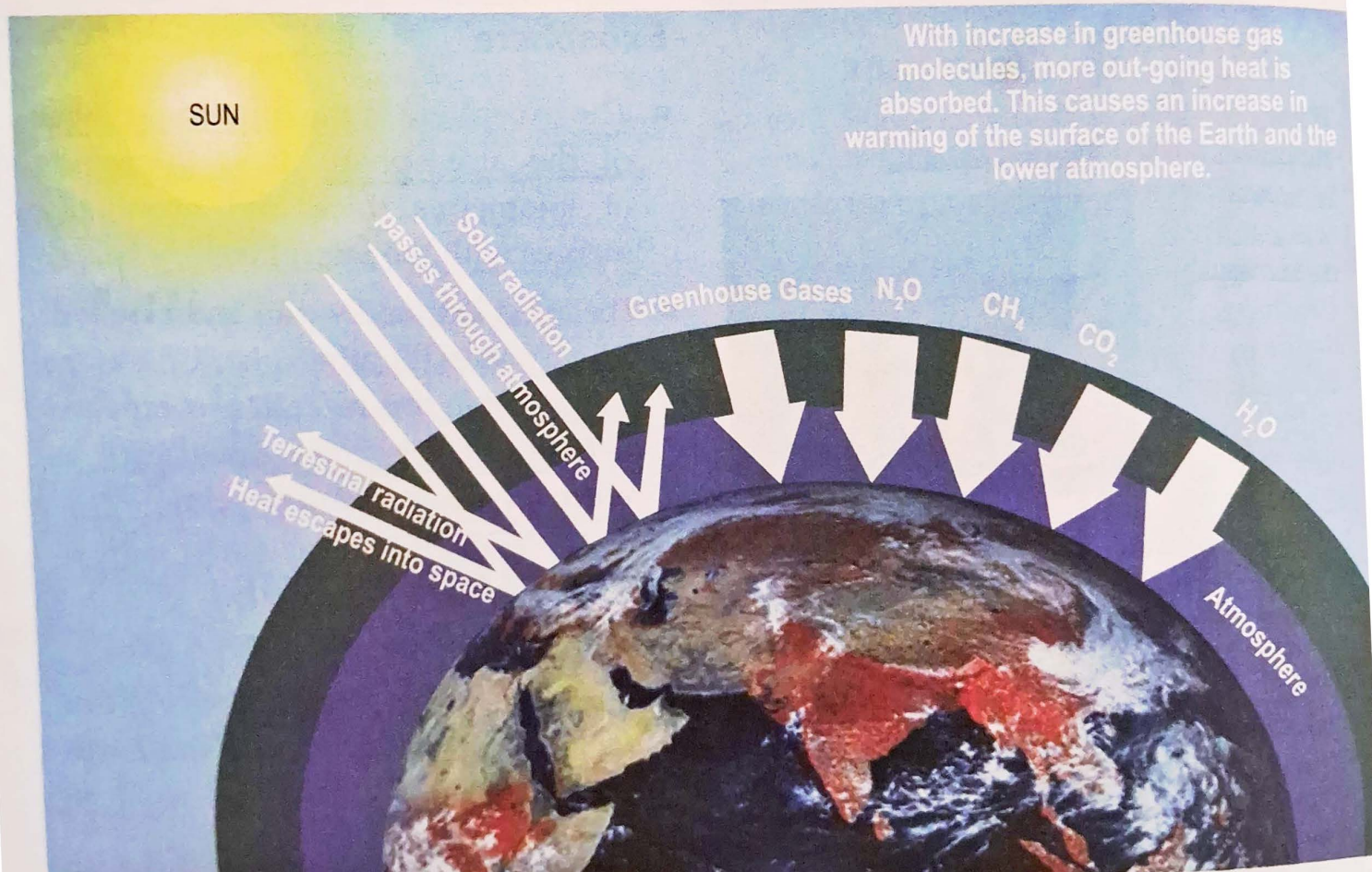


Fig. 2.3: the greenhouse effect

KEY WORDS

precipitation: The process in which the liquid and solid water particles that fall from clouds and reach the ground are known as precipitation. These particles include drizzle, rain, snow, ice crystals and hail of which rain is the most common.

greenhouse gases: these are those gases in the air that have the ability to trap more heat than the others. Common GHGs are water vapour, carbon dioxide and methane.

Over the years the heat-trapping gases called GHGs have increased in the atmosphere, thus making the planet warmer. In addition to carbon dioxide, carbon monoxide and water vapour, these gases also include methane, nitrous oxide and ozone. There are also some human-made GHGs such as CFCs and halons, which destroy the ozone layer and are called **Ozone Depleting Substances (ODS)**. A high concentration of GHGs and ODS traps the heat radiated from the Earth and increases the temperature of the atmosphere. It has led to a gradual rise in global temperatures, which is referred to as **global warming**.

Causes of Global Warming

Certain human activities have contributed to global warming.

1. **Burning of fossil fuel:** Amongst all the GHGs the significant contributor to the warming is carbon dioxide that is released in large proportions when fossil fuels such as coal and petroleum are burnt, whether in industries, power plants or by vehicular traffic. These changes have been noticed since the Industrial Revolution, when coal as a source of energy began to be used, later followed by petroleum. The table given on page 20 shows the sources of the major GHGs.
2. **Harmful emission from industries and vehicles:** Industries not only burn
- fossil fuel but also emit harmful gases such as oxides of nitrogen, sulphur and methane that pollute the air. Rapid urbanisation and industrialisation have increased industrial and vehicular pollution significantly.
3. **Deforestation and mining:** Due to rapid growth of population, large areas under forest cover are being cleared to give way to settlements, agriculture, mining activities or industries. Forests are a major carbon sink and in their absence the carbon dioxide level in the atmosphere increases, which traps heat in the atmosphere leading to global warming.
4. **Rearing of farm animals in large numbers:** Keeping a large number of farm animals such as goats, sheep and buffaloes also contributes to global warming. When farm animals digest their food, methane is produced in their stomach, which is released into the atmosphere through their dung. By removing large sections of forests to expand farmland and raising farm animals instead, humankind further disturbs the delicate balance in nature by increasing the amount of methane in the atmosphere.
5. **Fertilisers:** Artificial nitrogen is frequently used as a fertiliser for crops. Some of this nitrogen is released into the atmosphere as nitrogen oxide. Thus, fertilising the crops with

KEY WORD

carbon sink: a reservoir of carbon created on natural absorption of carbon dioxide from the atmosphere by plants during the process of photosynthesis

artificial nitrogen increases the levels of nitrogen oxide in the air.

(Sources of Greenhouse Gases)

CO ₂	<u>Burning of fossil fuels in industries and for transportation, deforestation</u>
Methane	<u>Natural wetlands, livestock wastes, paddy fields</u>
Nitrogen oxide	<u>Use of chemical fertilisers, burning of fossil fuels</u>
Ozone	<u>Transported from the ozone layer</u>
CFC	<u>Liquid coolants in refrigerators, aerosols and foam use CFC</u>

Impact of Global Warming

The destructive impact of global warming is observed in various spheres of life and the environment.

- High temperatures lead to higher evaporation rate and drying up of the soil and surface water. This affects crop production. The occurrence of droughts is aggravating the problem even further.
- The heat waves in summer lead to a greater number of deaths due to heat strokes. They also cause the spread of vector-borne diseases such as malaria

KEY WORD

vector-borne diseases: diseases spread by organisms, named vectors, such as mosquitoes, ticks or sand flies.

and dengue, as well as water-borne diseases such as cholera and typhoid.

- Forest fires become more frequent.



Forest fires like these have become very frequent now.

- Tropical cyclones and hurricanes become a common occurrence as rainfall patterns change.
- Melting of glaciers occurs, affecting permanent ice sheets in both the Arctic and Antarctic regions, as well



melting of glaciers increase the occurrence of floods

as the snow-capped mountains. This increases the volume of water in the oceans and may also result in flooding along the riverbanks.

KEY WORDS

climate refugees: people displaced from their home country owing to climate disasters such as cyclones and floods
ecosystems: an ecosystem includes all living things (plants, animals and organisms) in a given area, interacting with each other and also with their non-living environment (weather, the Earth, Sun, soil, climate and atmosphere). In an ecosystem, each organism has its own role to play.

- Owing to increase in the sea surface temperature, sea levels rise in coastal areas and cause submergence of several islands. This leads to ecological and cultural loss and also gives rise to the problem of climate refugees.
- Owing to rise in ocean temperature, coral reefs are destroyed. This greatly affects the marine ecosystems.
- It is also predicted that more than a million species will experience loss of habitat and gradually become endangered and extinct.

FACT

I Am a Geographer!



• initiative
• critical thinking

Climate Change Scoreboard

Create a climate change scoreboard for the class to identify and track different actions you can take at home and at school to help slow climate change. There should be a section on the scoreboard for each student in the class. Think of ways to save energy, reduce pollution or remove carbon dioxide from the atmosphere. First, add the activities you currently do. Keep on adding your ideas to the scoreboard. Everyone is free to adopt any other activities on the list too. Discuss barriers you might face in your efforts and consider ways to overcome them.

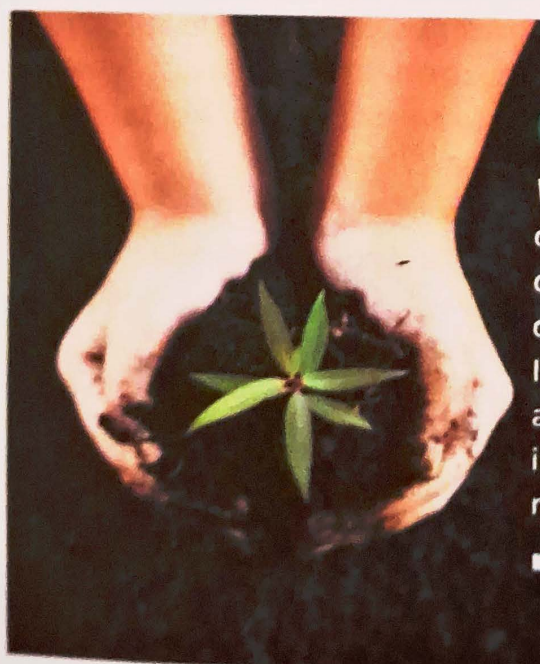
Track your actions for a week. To keep score, fill in a point each time you take an action during the week. At the end of the week, the students' scores can be added.

You should continue these activities after your class project ends. Continue to think of other conservation opportunities all through the year and put them into action.

Ways to Reduce Global Warming

To foster equilibrium in our environment and reduce the impact of global warming, certain measures need to be taken up worldwide. Some of these include the following:

1. We need to decrease emission of GHGs by reducing the burning of fossil fuels such as coal and petroleum. Instead, we should switch over to renewable sources of energy such as wind, Sun, water and tides. They are not only cheap and abundant but also non-polluting.
2. By planting more trees to increase forest cover, we can reduce excessive release of carbon dioxide in the air. The government also distributes free saplings and organises afforestation programmes to spread awareness regarding the beneficial effects of trees.
3. We should switch to eco-friendly cars and gadgets. For example, cars that run on electricity rather than fossil fuels are becoming popular all over the world; incandescent light bulbs should be replaced by Compact Fluorescent Lamps (CFL bulbs), because they consume less energy as compared to incandescent light bulbs.
4. We can save electricity and reduce global warming by turning off electrical gadgets such as lights, fans, air-conditioners, television and computer when we do not intend to use them.
5. Efforts should be made to hasten the development of green cities or eco-cities. These cities are urban areas around the world striving to lessen the environmental impacts of urbanisation by reducing waste, encouraging recycling, lowering emission levels as well as reducing levels of air, water



Curiosity Corner

Van Mahotsav, a week-long festival of tree planting is organised in India every year in the month of July. Thousands of trees are planted with the aim of increasing the green cover in the country. This annual event was launched in 1950. It was hoped that this event would create consciousness among the people about the importance of trees. However, indiscriminate felling of trees still continues in India and must be stopped.

- Do you think these newly planted trees will be able to contribute enough oxygen supply to balance the loss of old trees?

21st
CENTURY
SKILLS

- information literacy
- critical thinking
- problem-solving
- communication
- environmental awareness

and soil pollution. Some notable green cities include Oslo, Vancouver, Copenhagen, Amsterdam, Stockholm, Zurich, Essen, Sydney, Singapore and San Francisco.

6. By following the 3Rs—Reduce, Recycle and Reuse—strategy we can use natural resources judiciously for our growth as well as save them for the need of the future generations. This is called sustainable development.

- We must 'reduce' the use of natural resources so that they do not get depleted.
- We must 'reuse' resources instead of throwing them away after each use. Reuse conserves new resources and prevents old resources from entering the waste stream.
- We must 'recycle' or convert waste materials into new products, changing them from their original form by physical and chemical processes. In this way, not only is waste reduced but resources are also conserved.

THE OZONE LAYER AND ITS IMPORTANCE

Another major environmental concern is the depletion of the ozone layer in the stratosphere.

Ozone (O_3) is a heat-trapping gas. Ozone in the stratosphere acts as a shield, stopping the harmful ultraviolet (UV) radiation from reaching the Earth.

Causes of Depletion of Ozone Layer

Certain human activities release CFCs and other harmful chemicals that destroy the beneficial ozone layer. ODSs contain chlorine that strikes the ozone and splits

cross-curricular connect: Chemistry

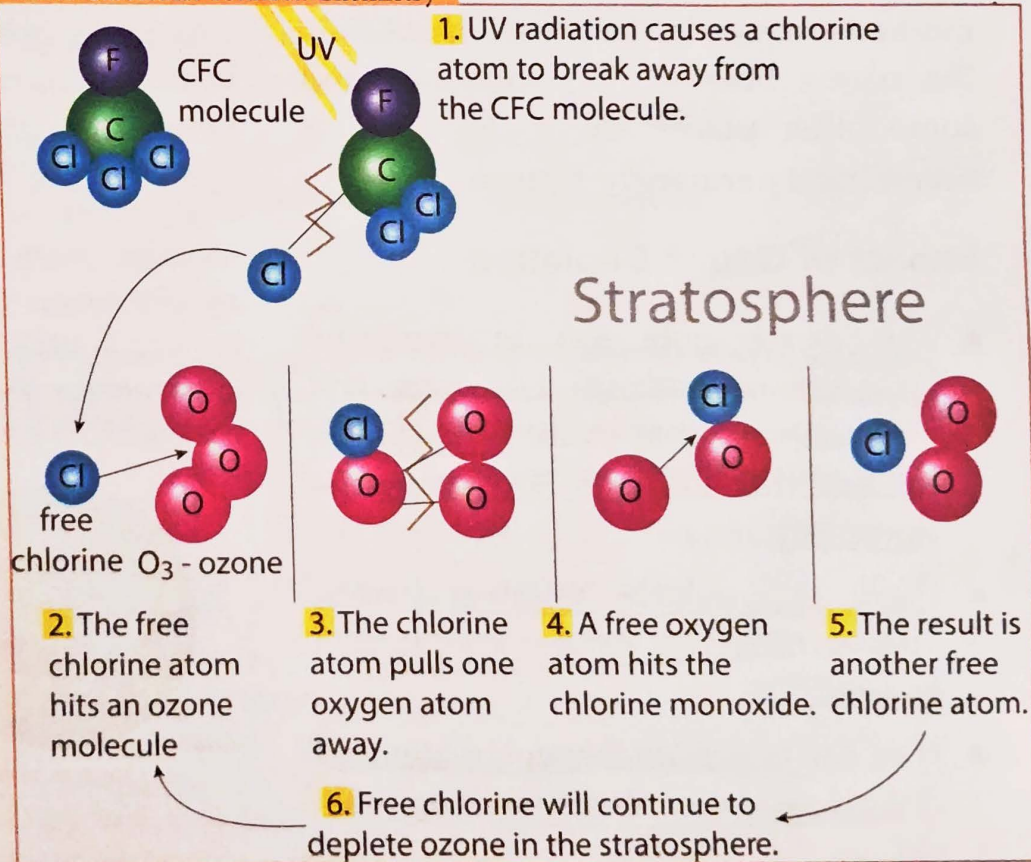


Fig. 2.4: How chlorine depletes ozone in the atmosphere.

it apart. Thus, they are a major cause behind the thinning of the ozone layer.

CFCs are far less abundant than carbon dioxide, but they are 10,000 times more powerful as a GHG and can remain in the atmosphere for more than 100 years. These ODS are released due to disposal of junk refrigerators, air conditioners and other electronic gadgets. Release of aerosols from spray cans, fire extinguishers, jet aircraft and solvent cleaners also contributes towards the generation of ODS. Volcanic eruptions also contribute to ozone depletion.

Although ozone depletion has been taking place for several years, the ozone layer was repairing itself naturally. With marked increase in the emission of CFCs, the ozone hole above Antarctica and some other places above the Earth is becoming a permanent feature.

Impact of Ozone Depletion

- The ozone hole lets in increased amounts of harmful UV rays and infrared rays that cause skin cancer, cataract in the eyes and suppress immunity.
- These rays adversely affect plants. This is observed from their stunted growth.
- They can penetrate through water and destroy aquatic plants and animals too.

KEY WORDS

immunity: the protection that someone's body gives them against a particular disease

polar vortex: it is a set of swirling cold winds around the North Pole that move in a circular motion.

Efforts to Prevent Ozone Depletion

1. Several countries met in the city of Montreal in Canada on 16 September 1987 to agree to the **Montreal Protocol**. The chief objective was to reach a global agreement to reduce the production of the ODS. As a result, the ozone layer is slowly recovering.
2. All over the world, 16 September is celebrated as the **World Ozone Day** to bring awareness about the preservation of the ozone layer.
3. We should purchase products that are labeled 'ozone friendly' or 'CFC-free'.

Did You Know?

The ozone hole above the Arctic, which opened in March 2020, mainly due to winds called **polar vortex**, has closed. The ozone hole over the Antarctic has also been shrinking 1–3 per cent per decade, since 2000. This has happened mainly due to the restrictions put on ODS since the signing of the Montreal Protocol in 1987. The warming of temperature in Antarctica also attributes to the partial healing of the ozone hole over the South Pole.

